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# Time-Dependent System Reliability

## COURSE NOTES

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**Renlong Wang**

Ph.D. Student

Department of Civil and Environmental Engineering

Rice University

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# Chapter 1

## Discrete Parameter Markov Chain

**Definition 1.0.1** (Stochastic Process). A stochastic process  $\mathbf{X} = \{X(t), t \in \mathcal{T}\}$  is a collection of random variables. That is, for each  $t$  in the index set  $\mathcal{T}$ ,  $X(t)$  is a random variable.

The index is often interpreted as *time* and, as a result, we refer to  $X(t)$  as the *state* of the process at time  $t$ . The *state space* of a stochastic process is defined as the set of all possible values that the random variables  $X(t)$  can assume. For instance,

- $\{X(t), t = 0, 1, \dots\}$  is a discrete-time stochastic process indexed by the nonnegative integers;
- $\{X(t), t \geq 0\}$  is a continuous-time stochastic process indexed by the nonnegative real numbers.

Thus, a stochastic process is a family of random variables that describes the system evolution through time of some (physical) process. This gives the conceptual foundation for *Time-Dependent System Reliability*.

*Markov Property* states that given present state, forecast of future is independent of past, i.e.,

$$P\{\text{Future} \mid \text{Present, Past}\} = P\{\text{Future} \mid \text{Present}\}.$$

Then, a stochastic process that follows the Markov property is called a *Markov process*.

**Definition 1.0.2** (Discrete-Parameter Markov Chain). Let  $\{X_n, n = 0, 1, \dots\}$  be a stochastic process that takes on a finite or countable number of possible values. Suppose that

$$P\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0\} = P\{X_{n+1} = j \mid X_n = i\}$$

for all states  $i_0, i_1, \dots, i, j$  and all  $n \geq 0$ , then stochastic process  $\{X_n, n = 0, 1, \dots\}$  is known as a discrete-parameter Markov chain (DPMC). In addition, if  $P\{X_{n+1} = j \mid X_n = i\} = p_{ij}$  for any  $n \geq 0$ , it is a homogeneous DPMC.

In homogeneous DPMC,  $p_{ij}$  represents the probability that the process will when in state  $i$ , next make a transition into state  $j$ , which is called *one-step transition probability*. Let  $\mathbf{P}$  denote the matrix of  $p_{ij}$ , so that the

transition probability matrix is

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ p_{i0} & p_{i1} & p_{i2} & \cdots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix},$$

where  $0 \leq p_{ij} \leq 1$  and  $\sum_j p_{ij} = 1$  for all  $i$ .

Define the  $n$ -step transition probability  $p_{ij}(n)$  as

$$p_{ij}(n) = P\{X_{n+k} = j | X_k = i\}, \quad n \geq 0, i, j \geq 0.$$

**Theorem 1.0.1** (Chapman-Kolmogorov Equations). The *Chapman-Kolmogorov Equations* are

$$p_{ij}(n+m) = \sum_{k=0}^{\infty} p_{ik}(n)p_{kj}(m) \quad \forall n, m \geq 0, \forall i, j. \quad (1.1)$$

*Proof.*

$$\begin{aligned} p_{ij}(n+m) &= P\{X_{n+m} = j | X_0 = i\} && \leftarrow \text{Definition} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k | X_0 = i\} && \leftarrow \text{Law of total probability} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\} && \leftarrow \text{Conditional probability} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k\} P\{X_n = k | X_0 = i\} && \leftarrow \text{Markov property} \\ &= \sum_{k=0}^{\infty} p_{ik}(n)p_{kj}(m). \end{aligned}$$

□

If we let  $\mathbf{P}(n)$  denote the matrix of  $n$ -step transition probability  $p_{ij}(n)$ , then

$$\mathbf{P}(n+m) = \mathbf{P}(n) \cdot \mathbf{P}(m), \quad (1.2)$$

in particular,

$$\mathbf{P}(2) = \mathbf{P}^2 \quad \mathbf{P}(n) = \mathbf{P}^n.$$

**Remark 1.0.1.** The matrix of  $n$ -step transition probabilities is obtained by multiplying the matrix of one-step transition probabilities by itself  $n - 1$  times.

Let  $p_i(n)$  denote the probability that a Markov chain is in state  $i$  at time  $n$ , i.e.,

$$p_i(n) = P\{X_n = i\}.$$

In vector notation, let  $\mathbf{p}(n) = (p_1(n), p_2(n), \dots, p_i(n), \dots)$ .

The state probabilities at any time step  $n$  can be obtained from a knowledge of the initial probability distribution and the matrix of transition probabilities. From the theorem of total probability,

$$p_i(n) = \sum_{\text{all } k} p_k(0) P\{X_n = i | X_0 = k\} = \sum_{\text{all } k} p_k(0) p_{ki}(n),$$

writing in matrix form becomes

$$\mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}(n) = \mathbf{p}(0)\mathbf{P}^n,$$

where  $\mathbf{p}(0)$  denotes the initial state distribution.

**Definition 1.0.3** (Steady-State Distribution). Let  $\mathbf{P}$  be the transition probability matrix of a DPMC, and let the vector  $\boldsymbol{\pi}$  be a probability distribution, whose element  $\pi_j$  denote the probability of being in state  $j$ , i.e.,

$$0 \leq \pi_j \leq 1, \quad \sum_{\text{all } j} \pi_j = 1.$$

Then  $\boldsymbol{\pi}$  is said to be steady-state distribution if and only if

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}.$$

**Remark 1.0.2** (Steady-State Distribution & Eigenproblem). Recall the eigenvalue problem

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

where  $\lambda$  is an eigenvalue of  $\mathbf{A}$  and  $\mathbf{x}$  is the corresponding right eigenvector.

Since a steady-state distribution satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi},$$

$\boldsymbol{\pi}$  is a left eigenvector of  $\mathbf{P}$  associated with the eigenvalue  $\lambda = 1$ .

Equivalently,

$$\mathbf{P}^\top \boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top.$$

Thus, the stationary distribution can be computed by solving

$$\begin{cases} \boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \\ \boldsymbol{\pi}\mathbf{e} = 1. \end{cases} \quad (1.3)$$

where  $\mathbf{e}$  is the vector of ones.

Define  $f_{ij}(n)$  for  $i \neq j$  as the probability that, starting from state  $i$ , the first passage to state  $j$  occurs in exactly  $n$  steps

$$f_{ij}(n) = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}, \quad n = 1, 2, \dots$$

**Proposition 1.0.1** (Relationship between  $f_{ij}(n)$  and  $p_{ij}(n)$ ). For  $i \neq j$ ,

$$f_{ij}(n) = p_{ij}(n) - \sum_{l=1}^{n-1} f_{ij}(l)p_{ij}(n-l). \quad (1.4)$$

When  $i = j$ ,

$$f_{ii}(n) = p_{ii}(n) - \sum_{l=1}^{n-1} f_{ii}(l)p_{ii}(n-l). \quad (1.5)$$

*Proof.* If the chain is in state  $j$  at step  $n$ , then the first visit to state  $j$  must occur at some step  $\ell \leq n$ . If the first visit occurs at step  $\ell$ , then the probability of this event is  $f_{ij}(\ell)$ . After reaching state  $j$ , the probability of being in state  $j$  again after the remaining  $n - \ell$  steps is  $p_{jj}(n - \ell)$ . Therefore,

$$\begin{aligned} p_{ij}(1) &= f_{ij}(1), \\ p_{ij}(2) &= f_{ij}(1)p_{jj}(1) + f_{ij}(2), \\ p_{ij}(3) &= f_{ij}(1)p_{jj}(2) + f_{ij}(2)p_{jj}(1) + f_{ij}(3), \\ &\vdots \\ p_{ij}(n) &= f_{ij}(n) + \sum_{\ell=1}^{n-1} f_{ij}(\ell)p_{jj}(n-\ell). \end{aligned}$$

Rearranging gives

$$f_{ij}(n) = p_{ij}(n) - \sum_{\ell=1}^{n-1} f_{ij}(\ell)p_{jj}(n-\ell).$$

The case  $i = j$  follows analogously. □

Proposition 1.0.1 essentially gives the following relationship:

Being in state  $j$  at step  $n = \sum_{\ell=1}^n (\text{first reaching } j \text{ at step } \ell) \times (\text{being in } j \text{ again after the remaining } n - \ell \text{ steps}),$

and it also gives a recursive way to compute  $f_{ij}(n)$  based on  $p_{ij}(n)$ .

**Remark 1.0.3** (How to Calculate  $p_{ij}(n)$  and  $f_{ij}(n)$ ). Use CK equation in matrix form for  $p_{ij}(n)$ , and then use the recursive equation (1.4) for  $f_{ij}(n)$ .

Let

$$T_j = \min\{n \geq 1 : X_n = j\}$$

be the first return time to  $j$ . Then, we can define the mean recurrence time  $m_{jj}$  of state  $j$  as

$$m_{jj} = \mathbb{E}[\text{Recurrence time}] = \mathbb{E}[T_j] = \sum_{n=1}^{\infty} n f_{jj}(T_j = n) = \sum_{n=1}^{\infty} n f_{jj}(n),$$

which is the average number of steps taken to return to state  $j$  for the first time after leaving it. Similarly, for  $i \neq j$ ,

we can define the mean first passage time

$$m_{ij} = \sum_{n=1}^{\infty} n f_{ij}(n).$$

The  $m_{ij}$  uniquely satisfy

$$m_{ij} = f_{ij}(1) + \sum_{k \neq j} p_{ik}(1 + m_{kj}) = 1 + \sum_{k \neq j} p_{ik} m_{kj},$$

since the process in state  $i$  either goes to state  $j$  in one step with probability  $p_{ij}$ , or goes first to some intermediate state  $k$  in one step with probability  $p_{ik}$  and then eventually on to  $j$  in an additional  $m_{kj}$  steps.

Define  $q_{ij}(m)$  as the probability that, starting from state  $i$ , state  $j$  is reached for the first time within  $m$  steps

$$q_{ij}(m) = \sum_{n=1}^m f_{ij}(n).$$

Equivalently,

$$q_{ij}(m) = P\{T_j \leq m \mid X_0 = i\},$$

where  $T_j$  denotes the first passage time to state  $j$ . Since  $q_{ij}(m)$  is the cumulative probability associated with the first passage probabilities  $f_{ij}(n)$ ,

$$f_{ij}(m) = q_{ij}(m) - q_{ij}(m-1).$$

Letting  $m \rightarrow \infty$ , we define

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n) = \lim_{m \rightarrow \infty} q_{ij}(m),$$

which is the probability that state  $j$  is ever reached from state  $i$ . When  $i = j$ ,

$$f_{jj} = \sum_{n=1}^{\infty} f_{jj}(n) = \lim_{m \rightarrow \infty} q_{jj}(m),$$

which is the probability that, starting from state  $j$ , the process eventually returns to state  $j$ .

**Proposition 1.0.2** (Relationship among  $p_{ij}(n)$ ,  $f_{ij}(n)$ , and  $q_{ij}(n)$ ). For  $i \neq j$  and  $n \geq 1$

$$q_{ij}(n) \geq p_{ij}(n) \geq f_{ij}(n).$$

*Proof.* We prove this directly from the definitions. Recall that

$$\begin{aligned} p_{ij}(n) &= P(X_n = j \mid X_0 = i), \\ f_{ij}(n) &= P(X_n = j, X_k \neq j, k = 1, \dots, n-1 \mid X_0 = i), \\ q_{ij}(n) &= P(\min\{m : X_m = j\} \leq n \mid X_0 = i). \end{aligned}$$

First, split  $q_{ij}(n)$  according to whether the first visit to  $j$  occurs before time  $n$  or exactly at time  $n$ :

$$q_{ij}(n) = P(\min\{m : X_m = j\} \leq n-1 \mid X_0 = i) + \underbrace{P(\min\{m : X_m = j\} = n \mid X_0 = i)}_{f_{ij}(n)}.$$

Next, split  $p_{ij}(n)$  according to whether the chain visits  $j$  for the first time at time  $n$ , or has already visited  $j$  by

time  $n - 1$ :

$$p_{ij}(n) = P(X_n = j \mid X_0 = i) = \underbrace{P(\min\{m : X_m = j\} = n \mid X_0 = i)}_{f_{ij}(n)} + P(\min\{m : X_m = j\} \leq n - 1, X_n = j \mid X_0 = i).$$

Since

$$\{\min\{m : X_m = j\} \leq n - 1, X_n = j\} \subseteq \{\min\{m : X_m = j\} \leq n - 1\},$$

we have

$$P(\min\{m : X_m = j\} \leq n - 1, X_n = j \mid X_0 = i) \leq P(\min\{m : X_m = j\} \leq n - 1 \mid X_0 = i).$$

Therefore, comparing the two decompositions above gives

$$p_{ij}(n) \leq q_{ij}(n).$$

Also, from

$$p_{ij}(n) = f_{ij}(n) + P(\min\{m : X_m = j\} \leq n - 1, X_n = j \mid X_0 = i),$$

and the nonnegativity of probability, we obtain

$$p_{ij}(n) \geq f_{ij}(n).$$

Hence,

$$q_{ij}(n) \geq p_{ij}(n) \geq f_{ij}(n).$$

□

Let

$$L_j = \min\{n \geq 1 : X_n \neq j\}$$

be starting from state  $j$ , the number of steps until the process first leaves state  $j$ . Then, we can define the mean occupation time  $l_{jj}$  of state  $j$  as

$$l_{jj} = \mathbb{E}[\text{Occupation time}] = \mathbb{E}[L_j] = \sum_{n=1}^{\infty} n p_{jj}^{n-1} (1 - p_{jj}) = \frac{1}{1 - p_{jj}}.$$

**Proposition 1.0.3** (Closed-Form Expression of Mean Occupation Time). The closed-form of mean occupation time of state  $j$  is

$$l_{jj} = \mathbb{E}[L_j] = \frac{1}{1 - p_{jj}}.$$

*Proof.* Note that  $L_j \sim \text{Geom}(1 - p_{jj})$  and

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}, \quad |x| < 1.$$

Differentiate both sides with respect to  $x$  to obtain

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1 - x)^2}.$$

Letting  $x = p_{jj}$ ,

$$\sum_{n=1}^{\infty} np_{jj}^{n-1} = \frac{1}{(1-p_{jj})^2}.$$

Therefore,

$$l_{jj} = \sum_{n=1}^{\infty} np_{jj}^{n-1}(1-p_{jj}) = (1-p_{jj}) \sum_{n=1}^{\infty} np_{jj}^{n-1} = (1-p_{jj}) \frac{1}{(1-p_{jj})^2} = \frac{1}{1-p_{jj}}.$$

□